

Linear Programming for Lunar Base Power Allocation: An Artemis Foundation Habitat Project

Brendan Velasquez

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Abstract

This project models lunar base power allocation as a continuous linear program for a representative Artemis Foundation Surface Habitat scenario at the lunar south pole. The model uses two operating modes, crewed and dormant, and allocates limited source-side power across habitat, rover, science, and in-situ resource utilization (ISRU) loads using a weighted-shortage objective. Three nested datasets are tested: a small habitat/science case, a medium case with a remote rover, and a large case with split ISRU loads and delivery losses. The model is solved with custom tableau simplex and revised simplex implementations and with a PuLP/CBC solver baseline. All three methods return optimal solutions with compatible objective values and the same load-priority pattern. The small and medium datasets have zero shortage. In the large dataset, the crewed source-capacity constraint binds and produces objective value approximately 2.39202×10^5 . Habitat, rover, science, and dormant-mode ISRU loads are fully served, while crewed ridge-side ISRU is partially served and crewed crater-side ISRU is fully curtailed.

I. Introduction

Electrical power is a binding resource for sustained lunar surface operations. A crewed South Pole outpost must support habitat survival, crew operations, mobility, science activity, and eventually in-situ resource utilization under a limited generation architecture. Because these loads compete for finite source-side power, the allocation problem can be modeled as a linear optimization problem.

The scenario in this report is a representative Artemis South Pole habitat case with two operating modes and a limited source-side power architecture. The scenario in this project combines publicly available NASA habitat, logistics, fission-power, solar-array, and lunar-power reference values into a continuous LP for comparing custom simplex implementations with a standard solver. To keep the model linear, this project does not cover nonlinear storage dynamics or realistic electrical-network physics.

The computational part of the project has two roles. First, the custom tableau simplex and revised simplex codes are tested on small diagnostic LPs to verify that the general-form solver pipeline works:

it accepts inequality constraints, converts them to standard form, uses a Phase I auxiliary problem to find an initial feasible basis when artificial variables are needed, recovers an optimal solution when the model is feasible, and detects infeasibility when the model is inconsistent. Second, the same simplex algorithms are applied to the lunar power-allocation datasets and compared with a PuLP/CBC solver baseline. Basically, the small verification examples are meant to validate the algorithmic before it is used on the project model.

The first operating mode represents crewed operations during a nominal 30-day mission. The second represents dormant / keep-alive operations between crewed missions. The model uses fixed remote-delivery efficiencies and a weighted-shortage objective, so the optimization question is how to distribute source-side power while protecting high-priority services.

The remainder of the paper is organized as follows. Section II defines the mission scenario and research context. Section III presents the data, source status, and assumptions. Section IV formulates the linear program. Section V describes the custom simplex

pipeline, the algorithm-verification examples, and the lunar benchmark implementation. Section VI defines the datasets and experiments. Section VII reports the computational results. The final sections discuss interpretation, limitations, future work, and conclusions.

II. Mission scenario and research context

Mission phase. The model uses a Foundation Surface Habitat / Foundational Exploration-inspired scenario. NASA’s Moon-to-Mars Architecture describes a Foundational Exploration segment with increased-duration surface missions, expanded mobility, and regional exploration of the lunar South Pole [1]. Kessler et al. describe the Surface Habitat as an initial lunar surface habitat concept with 30-to-60-day habitability for a crew of up to four [2]. Csank et al. use the term Foundation Surface Habitat in a lunar-power context and list it as a primary asset for sustained lunar presence [6]. These sources motivate the scenario modeled here.

Crew and operating cycle. The baseline uses four crew as a modeling choice within the cited habitat range. The nominal crewed duration is 30 days, consistent with the lower end of the Surface Habitat concept and with Artemis Base Camp logistics references [2, 3]. The dormant mode represents the long uncrewed interval between crewed visits. Owens et al. describe an Artemis Base Camp planning case with approximately 30-day annual crewed missions and approximately 335 days of uncrewed dormant period between missions [3].

Site and environment. The lunar south pole is used because it is central to the cited Artemis regional exploration and lunar-power references. The site is also tied to water-ice prospecting and to power-system studies for south-pole missions [1, 5, 6, 7]. This report uses two operating modes and fixed source capacities to represent the higher-activity crewed regime and the lower-activity keep-alive regime.

Power architecture. The source-side architecture combines a 40 kW fission surface power reference with

a 10 kW solar-array reference to form a 50 kW crewed-mode case. NASA’s Fission Surface Power page describes a 40 kW-class fission system for lunar operation, and Pappa et al. describe a relocatable 10 kW solar array concept for lunar south-pole missions [4, 5]. The combined 50 kW crewed capacity is therefore a source-informed modeling choice. The dormant mode uses only the 40 kW fission reference.

III. Data, assumptions, and parameter selection

Source-side parameters. Table 1 summarizes the source-side architecture used in the model. The column labeled “source status” distinguishes direct source support from educated modeling assumptions.

Table 1: Power-source parameters.

Source	Symbol	Adopted value	Source status
Fission surface power	P_{fis}	40 kW	directly source-backed [4, 7]
Solar supplement	P_{sol}	10 kW	directly source-backed; crewed-only use is assumed [5, 7]
Storage / reserve	—	not modeled separately	omitted modeling feature

The total source-side capacities are

$$P_c = 50 \text{ kW}, \quad P_d = 40 \text{ kW}.$$

Because 2 kW of habitat keep-alive power is treated as a hard fixed load in each mode, the flexible-load capacities used in the LP are

$$R_c = P_c - 2 = 48, \quad R_d = P_d - 2 = 38.$$

This distinction between total source power P_m and flexible-load capacity R_m is used throughout the formulation.

Load inventory. Table 2 lists the subsystem classes used in the largest model. Habitat power values are based primarily on the Foundation Surface Habitat and lunar habitat power references [6, 7]. Rover, science, and ISRU values are taken or synthesized from the same lunar-power source family [6, 7].

Table 2: Subsystem/load inventory used in the full model.

Subsystem	Symbol	Adopted demand	Source status	Notes
Habitat survival / keep-alive	h_0	2 kW in both modes	directly source-backed	hard fixed load removed before flexible allocation
Habitat crew operations	h_1	18 kW crewed, 0 dormant	derived	20 kW crewed habitat total minus 2 kW keep-alive
Local rover / mobility	r_1	0.5 kW crewed, 0 dormant	representative value	within cited rover/science scale
Remote rover / mobility	r_2	0.5 kW crewed, 0 dormant	representative value	same delivered load as local rover, with delivery loss
Science payload	s	0.2 kW crewed, 0 dormant	source-informed	PRIME-1-scale fixed science load
Ridge-side ISRU	i_r	46 kW crewed, 8 kW dormant	crewed value source-backed; dormant split assumed	ridge-side processing load
Crater-side ISRU	i_c	22 kW crewed, 4 kW dormant	crewed value source-backed; dormant split assumed	crater-side extraction load

Table 3: Source status of numerical parameters used in the LP.

Parameter	Value used	Source status	Citation / modeling note
Fission source	40 kW	directly source-backed	NASA Fission Surface Power; Csank 2025 [4, 7]
Solar supplement	10 kW	directly source-backed	Pappa RSA; Csank VSAT [5, 7]
Crewed habitat total	20 kW	directly source-backed	Csank 2022 and 2025 [6, 7]
Habitat keep-alive	2 kW	directly source-backed	Csank 2022 and 2025 [6, 7]
Habitat operations increment	18 kW	derived	$20 - 2 = 18$
Rover loads	0.5 kW each	source-informed representative value	VIPER / rover scale [6, 7]
Science payload	0.2 kW	source-informed value	PRIME-1-scale payload [6]
Crewed ISRU ridge/crater	46 kW / 22 kW	directly source-backed	Csank 2022 [6]
Dormant ISRU total	12 kW	directly source-backed	Csank 2025 [7]
Dormant ISRU ridge/crater	8 kW / 4 kW	modeling split	split of the 12 kW keep-alive value
Remote efficiencies	0.95, 0.92, 0.90	modeling assumption	fixed sensitivity parameters, not NASA-reported efficiencies
Dormant solar availability	0 kW	modeling assumption	crewed-only solar supplement

Transmission and delivery assumptions. The model uses fixed delivery efficiencies for remote loads for simplicity. The cited lunar-power studies support the relevance of remote delivery and kilometer-scale separation for ISRU and power assets, but they do not provide the exact efficiencies used here [6, 7]. The values 0.95, 0.92, and 0.90 seen below were chosen as simple modeling assumptions to represent mild-to-moderate delivery losses, with larger losses assigned to the more remote ISRU loads. Thus, the remote efficiencies are modeling parameters:

$$\eta_{r_2} = 0.95, \quad \eta_{i_r} = 0.92, \quad \eta_{i_c} = 0.90.$$

All local loads use unit efficiency.

Penalty weights. The shortage-penalty weights are modeling choices for the priority order used by the LP, they are not official external NASA risk weights. Habitat operations receive the largest weight, mobility receives the next largest weight, the science payload is lower, and ISRU is lowest because this model treats ISRU as deferrable relative to immediate habitat and mobility service.

IV. Linear programming formulation

Sets and indices. For operating modes, let

$$\mathcal{M} = \{c, d\},$$

where c denotes crewed mode and d denotes dormant mode. For the full subsystem set, let

$$\mathcal{I} = \{h_0, h_1, r_1, r_2, s, i_r, i_c\}.$$

The fixed habitat keep-alive load h_0 is removed before optimizing the flexible loads, so define

$$\mathcal{F} = \{h_1, r_1, r_2, s, i_r, i_c\}.$$

For experiments, the dataset index is

$$\mathcal{D} = \{\text{small}, \text{medium}, \text{large}\}.$$

Decision variables. For each $i \in \mathcal{F}$ and $m \in \mathcal{M}$, define

$y_{i,m}$ = source-side power dispatched to i
in mode m ,

$x_{i,m}$ = delivered power received by i
in mode m ,

$s_{i,m}$ = unmet delivered demand for i
in mode m .

All decision variables are continuous and nonnegative.

Parameters. The operating-mode durations are

$$\tau_c = 30 \cdot 24 = 720, \quad \tau_d = 335 \cdot 24 = 8040$$

hours. The flexible-load capacities are

$$R_c = 48, \quad R_d = 38.$$

The delivered-demand vectors for the flexible loads are

$$D_c = \begin{pmatrix} 18 \\ 0.5 \\ 0.5 \\ 0.2 \\ 46 \\ 22 \end{pmatrix}, \quad D_d = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 8 \\ 4 \end{pmatrix},$$

with flexible-load order

$$(h_1, r_1, r_2, s, i_r, i_c).$$

The shortage-penalty vector is

$$p = \begin{pmatrix} 100 \\ 25 \\ 25 \\ 10 \\ 8 \\ 8 \end{pmatrix}.$$

Objective. The model minimizes weighted unmet delivered power over the two operating modes:

$$\min \tau_c \sum_{i \in \mathcal{F}} p_i s_{i,c} + \tau_d \sum_{i \in \mathcal{F}} p_i s_{i,d}.$$

Because shortages are measured in kW and multiplied by mode duration, the objective has units of weighted kWh-shortage.

Constraints. The full model is subject to the following constraints.

Fixed habitat preprocessing. The hard keep-alive load is not optimized; it is fixed at 2 kW in both modes and has unit efficiency, so it is subtracted from the total source power before the flexible-load LP is solved. Thus

$$R_c = 50 - 2 = 48, \quad R_d = 40 - 2 = 38.$$

Source-capacity constraints.

$$\sum_{i \in \mathcal{F}} y_{i,c} \leq R_c = 48, \quad \sum_{i \in \mathcal{F}} y_{i,d} \leq R_d = 38.$$

Delivery-efficiency relations.

$$x_{i,m} = \eta_i y_{i,m}, \quad \forall i \in \mathcal{F}, \forall m \in \mathcal{M}.$$

Demand-balance constraints.

$$x_{i,m} + s_{i,m} = D_{i,m}, \quad \forall i \in \mathcal{F}, \forall m \in \mathcal{M}.$$

Nonnegativity.

$$y_{i,m} \geq 0, \quad x_{i,m} \geq 0, \quad s_{i,m} \geq 0, \quad \forall i \in \mathcal{F}, \forall m \in \mathcal{M}.$$

Final LP model statement.

$$\begin{aligned}
\min \quad & \tau_c \sum_{i \in \mathcal{F}} p_i s_{i,c} + \tau_d \sum_{i \in \mathcal{F}} p_i s_{i,d} \\
\text{s.t.} \quad & \sum_{i \in \mathcal{F}} y_{i,c} \leq 48, \\
& \sum_{i \in \mathcal{F}} y_{i,d} \leq 38, \\
& x_{i,m} = \eta_i y_{i,m}, \quad \forall i \in \mathcal{F}, \forall m \in \mathcal{M}, \\
& x_{i,m} + s_{i,m} = D_{i,m}, \quad \forall i \in \mathcal{F}, \forall m \in \mathcal{M}, \\
& y_{i,m} \geq 0, \quad x_{i,m} \geq 0, \quad s_{i,m} \geq 0, \\
& \quad \forall i \in \mathcal{F}, \forall m \in \mathcal{M}.
\end{aligned}$$

Interpretation. The LP minimizes weighted service shortfall across the operating cycle represented by the two modes. Habitat survival is enforced before the optimization by subtracting the fixed keep-alive load. The remaining loads may be partially curtailed if source-side power is insufficient. The crewed mode is the tighter regime because it includes habitat activity, mobility, science, and optional high-power ISRU operation. The dormant mode contains only keep-alive and ISRU maintenance loads.

V. Implementation and computational setup

Overview of the custom simplex pipeline. The submitted script contains two custom simplex implementations: a tableau simplex method and a revised simplex method. Both are written for minimization problems with nonnegative decision variables. The general input form is

$$\min c^T x$$

subject to linear constraints of the form

$$a_k^T x \leq b_k, \quad a_k^T x = b_k, \quad \text{or} \quad a_k^T x \geq b_k,$$

with

$$x \geq 0.$$

The preprocessing routine records the constraint sense for each row, normalizes rows with negative right-hand side values, and then converts the model to equality form. For $\leq$ constraints it adds slack variables. For $\geq$ constraints it adds surplus variables and

artificial variables. For equality constraints it adds artificial variables when no obvious starting basic variable is available.

Two-Phase initialization. The Phase I problem minimizes the sum of the artificial variables introduced during preprocessing. If the Phase I optimum is positive beyond the numerical tolerance, the original LP is declared infeasible. If the Phase I optimum is zero, the code removes artificial variables when possible, recovers a feasible basis for the original constraints, and then starts Phase II with the original objective. This step is what allows the same simplex code to handle general-form LPs rather than only already-prepared standard-form inputs.

Tableau simplex implementation. The tableau implementation stores the active Phase I or Phase II problem as a simplex tableau and performs pivot operations directly on that tableau. At each iteration, the entering variable is selected using the reduced-cost optimality test for a minimization problem, with Bland-style tie-breaking to avoid cycling. The leaving row is selected by the ratio test, again using a Bland-style tie-break if several rows attain the same minimum ratio. The output records the final status, objective value, solution values for the original variables, pivot count, standard-form size, and runtime.

Revised simplex implementation. The revised simplex implementation uses the same preprocessing, Phase I construction, and tie-breaking logic, but it does not update the full tableau. Instead, it stores a basis and computes the needed quantities from the basis matrix. For a current basis B , it computes

$$x_B = B^{-1}b$$

and reduced costs of the form

$$\bar{c}_j = c_j - c_B^T B^{-1} A_j.$$

If every reduced cost is nonnegative, the current basic feasible solution is optimal for the minimization problem. If some reduced cost is negative, the entering column is chosen, the direction

$$u = B^{-1} A_j$$

is computed, and the leaving variable is chosen by the ratio test. Thus the tableau and revised implementations solve the same standard-form LPs, but they store and update the simplex information differently.

Algorithm verification examples. Before applying the custom simplex routines to the lunar datasets, the script checks them on three small diagnostic LPs. The first diagnostic problem is the feasible lower-bound LP

$$\min 3x_1 + 4x_2$$

subject to

$$2x_1 + 3x_2 \geq 13, \quad x_1 \geq 2, \quad x_2 \geq 5, \quad x_1, x_2 \geq 0.$$

The expected optimum is

$$x_1 = 2, \quad x_2 = 5, \quad 3x_1 + 4x_2 = 26.$$

This example checks that the code can process $\geq$ -constraints, introduce surplus and artificial variables, run Phase I, remove artificial variables after feasibility is recovered, and return the correct Phase II optimum.

The second diagnostic problem is the mixed-sense feasible LP

$$\min x_1 + 2x_2$$

subject to

$$x_1 + x_2 \leq 6, \quad x_1 - x_2 = 1, \quad x_1 \geq 3, \quad x_1, x_2 \geq 0.$$

The expected optimum is

$$x_1 = 3, \quad x_2 = 2, \quad x_1 + 2x_2 = 7.$$

This example checks that the same general-form preprocessing routine can handle a feasible LP containing $\leq$, $=$, and $\geq$ constraints in one model.

The third diagnostic problem is the infeasible bound LP

$$\min x_1$$

subject to

$$x_1 \geq 2, \quad x_1 \leq 1, \quad x_1 \geq 0.$$

This example checks that Phase I correctly identifies an inconsistent constraint system before Phase II is attempted. Table 4 reports the script output for these verification runs, and the complete console output is reproduced in Appendix A.

Connection to the lunar benchmark. The diagnostic examples test the general-form machinery: constraint-sense handling, standard-form conversion, artificial-variable initialization, Phase I feasibility recovery, mixed-sense feasible input, and infeasibility detection. The lunar benchmark then uses the same tableau and revised simplex kernels on structured standard-form matrices built from the LP in Section IV. The lunar model does not require Phase I because the shortage variables and source-capacity slacks provide an immediate basic feasible solution. Thus the diagnostic examples test the difficult initialization cases, while the lunar datasets test the algorithms on the project application and provide the runtime comparison with PuLP/CBC.

Lunar benchmark standard form. The lunar benchmark is built from the LP in Section IV. For a dataset with n flexible loads, the script keeps source-side dispatch variables, delivered-power variables, shortage variables, and two source-capacity slack variables. Thus the implemented benchmark standard form has

$$6n + 2$$

variables and

$$4n + 2$$

constraints. For the large dataset, $n = 6$, so the benchmark matrix has 38 variables and 26 constraints. This full benchmark form is the one used for the computational tables.

Initial BFS for the lunar benchmark. The lunar application has a direct initial BFS after fixed habitat preprocessing. For the implemented full standard form, the initial point is

$$y_{i,m} = 0, \quad x_{i,m} = 0, \quad s_{i,m} = D_{i,m},$$

for each flexible load and mode, together with capacity slacks

$$u_c = 48, \quad u_d = 38.$$

This basis is feasible because every demand-balance equation is satisfied by the shortage variable and every source-capacity equation is satisfied by the corresponding capacity slack.

Table 4: Verification of the custom general-form simplex pipeline before the lunar benchmark.

Verification problem	Method	Status	Reported result	Objective	Phase I / II pivots	Std. size	Runtime
Feasible lower-bound LP	Tableau	optimal	$x_1 = 2, x_2 = 5$	26	3 / 0	5 vars, 3 cons	0.000333 s
Feasible lower-bound LP	Revised	optimal	$x_1 = 2, x_2 = 5$	26	3 / 0	5 vars, 3 cons	0.000262 s
Mixed-sense feasible LP	Tableau	optimal	$x_1 = 3, x_2 = 2$	7	2 / 0	4 vars, 3 cons	0.000128 s
Mixed-sense feasible LP	Revised	optimal	$x_1 = 3, x_2 = 2$	7	2 / 0	4 vars, 3 cons	0.000122 s
Infeasible bound LP	Tableau	infeasible	detected in Phase I	—	1 / 0	checked in Phase I	0.000052 s
Infeasible bound LP	Revised	infeasible	detected in Phase I	—	1 / 0	checked in Phase I	0.000057 s

Reduced crewed subproblem for interpretation.

The following reduced crewed subproblem is not the exact matrix used by the benchmark script. It is an analytical simplification used to explain why the large-case allocation has the reported structure. For the crewed large dataset, delivered-power variables can be eliminated from the application LP by substituting

$$x_i = \eta_i y_i.$$

This gives the equivalent reduced crewed subproblem

$$\min 720 \left(100s_{h_1} + 25s_{r_1} + 25s_{r_2} + 10s_s + 8s_{i_r} + 8s_{i_c} \right)$$

subject to

$$\begin{aligned} y_{h_1} + y_{r_1} + y_{r_2} + y_s + y_{i_r} + y_{i_c} + u_c &= 48, \\ y_{h_1} + s_{h_1} &= 18, \\ y_{r_1} + s_{r_1} &= 0.5, \\ 0.95y_{r_2} + s_{r_2} &= 0.5, \\ y_s + s_s &= 0.2, \\ 0.92y_{i_r} + s_{i_r} &= 46, \\ 0.90y_{i_c} + s_{i_c} &= 22, \end{aligned}$$

with

$$y_i \geq 0, \quad s_i \geq 0, \quad u_c \geq 0.$$

The reduced crewed variable vector is

$$z_c = (y_{h_1}, y_{r_1}, y_{r_2}, y_s, y_{i_r}, y_{i_c}, s_{h_1}, s_{r_1}, s_{r_2}, s_s, s_{i_r}, s_{i_c}, u_c)^T \in \mathbb{R}_{\geq 0}^{13}.$$

The dormant reduced system is obtained by replacing the right-hand sides with

$$38, 0, 0, 0, 0, 8, 4.$$

From the demand equations of the reduced crewed large subproblem,

$$s_i = D_i - \eta_i y_i,$$

so minimizing weighted shortage is equivalent to maximizing protected weighted service,

$$\max \sum_{i \in \mathcal{F}} 720 p_i \eta_i y_i$$

subject to

$$\sum_{i \in \mathcal{F}} y_i \leq 48, \quad 0 \leq y_i \leq \frac{D_i}{\eta_i}.$$

The crewed weighted-service coefficients are

$$100, \quad 25, \quad 23.75, \quad 10, \quad 7.36, \quad 7.20,$$

in the order

$$(h_1, r_1, r_2, s, i_r, i_c).$$

Therefore the improving allocation order is

$$h_1 \rightarrow r_1 \rightarrow r_2 \rightarrow s \rightarrow i_r \rightarrow i_c.$$

The first four flexible loads are fully served:

$$\begin{aligned} y_{h_1} &= 18, & y_{r_1} &= 0.5, \\ y_{r_2} &= \frac{0.5}{0.95} = 0.52631579, & y_s &= 0.2. \end{aligned}$$

After these assignments, the remaining crewed source-side power is

$$48 - 18 - 0.5 - 0.52631579 - 0.2 = 28.77368421.$$

So the remaining power goes next to ridge-side ISRU:

$$y_{i_r} = 28.77368421, \quad y_{i_c} = 0.$$

Then

$$\begin{aligned} x_{i_r} &= 0.92(28.77368421) = 26.47178947, \\ s_{i_r} &= 46 - 26.47178947 = 19.52821053, \end{aligned}$$

and

$$\begin{aligned} x_{i_c} &= 0, \\ s_{i_c} &= 22. \end{aligned}$$

These values are reported again in the computational results.

Standard solver comparison. The solver baseline was implemented in PuLP and solved with CBC. The PuLP model uses the model-level variables from Section IV, while the custom simplex routines solve the standard-form matrices described above. The solver-level variable counts and simplex variable counts therefore differ. The custom methods solve preassembled matrices, while PuLP/CBC includes modeling-interface and solver-startup overhead. The runtime comparison should just be read as a small-instance implementation comparison.

Runtime protocol. For each dataset, the computational phase records runtime, objective value, number of decision variables, number of constraints, iteration count when applicable, solver / simplex status, capacity slack, and main allocation values. The solver runtimes in Section VII are single-run clock times from PuLP/CBC. The tableau and revised runtimes are single-run clock times from the custom simplex implementations.

VI. Datasets and experiments

Dataset design. The computational study uses three nested datasets. The small case isolates habitat operations, one local rover, and a science payload. The medium case adds a remote rover load and its associated efficiency loss. The large case is the full model used in this report, including split ISRU loads and fixed remote-delivery efficiencies. The progression increases the number of flexible loads from 3 to 6, which increases both the general LP size and the standard-form matrix size.

Experiment list. For each dataset, the computational phase performs the following experiments:

1. solve with tableau simplex,
2. solve with revised simplex,
3. solve with PuLP/CBC,
4. record runtime and model size,
5. compare objective values, capacity slack, and allocation values.

VII. Results

Optimal allocations. The PuLP/CBC solver, the tableau simplex implementation, and the revised simplex implementation all found optimal solutions for all three datasets. For the small and medium datasets, the optimal objective value is 0, so all modeled flexible demands are fully served. In the medium case, the remote rover receives 0.5 delivered kW from 0.52631579 dispatched kW because of the 0.95 delivery efficiency.

The large dataset is the first time resources have become a little scarce. In crewed mode, habitat operations, the local rover, the remote rover, and the science payload are fully served. The remaining flexible source-side power goes to ridge-side ISRU. Crewed ridge-side ISRU receives about 26.47 delivered kW and has shortage about 19.53 kW. Crewed crater-side ISRU receives 0 delivered kW and has shortage 22 kW. In dormant mode, both ISRU keep-alive loads are fully served. Table 6 gives the large-case load-level allocation values from the custom simplex runs.

The dormant large case can be checked directly from the LP data:

$$\begin{aligned} \frac{8}{0.92} + \frac{4}{0.90} &= 8.69565217 + 4.44444444 = \\ &13.14009661 < 38. \end{aligned}$$

So the dormant capacity slack is

$$u_d = 38 - 13.14009661 = 24.85990339,$$

which matches the tableau and revised outputs up to rounding. The large-case delivered allocation and shortage pattern is summarized in Figure 2, which is collected at the end of this report.

Table 5: Dataset structure and model dimensions. The report load count includes the fixed habitat keep-alive load h_0 ; the flexible-load list gives the optimized loads.

Case	Flexible loads	Report load count	General LP variables	Standard-form variables	Constraints
Small	h_1, r_1, s	4	18	20	14
Medium	h_1, r_1, r_2, s	5	24	26	18
Large	$h_1, r_1, r_2, s, i_r, i_c$	7	36	38	26

Table 6: Large-case load-level allocation from the tableau and revised simplex solutions. Values are in kW.

Load	Crewed dispatch	Crewed delivered	Crewed shortage	Dormant dispatch	Dormant delivered	Dormant shortage
h_1	18.000	18.000	0.000	0.000	0.000	0.000
r_1	0.500	0.500	0.000	0.000	0.000	0.000
r_2	0.526	0.500	0.000	0.000	0.000	0.000
s	0.200	0.200	0.000	0.000	0.000	0.000
i_r	28.774	26.472	19.528	8.696	8.000	0.000
i_c	0.000	0.000	22.000	4.444	4.000	0.000

Objective values. Table 7 reports objective values and iteration counts. The large-case PuLP/CBC objective is 239202.49536, while the tableau and revised objectives are 239202.49263. The absolute difference is about 2.73×10^{-3} , which is negligible relative to the 2.39×10^5 objective scale. The discrepancy is consistent with solver tolerance and printed rounding. The same load-priority pattern is returned across the three methods.

Runtime comparison. Table 8 reports the runtime and model-size comparison. For PuLP/CBC, the measured runtimes were approximately 0.01176 seconds for the small case, 0.01088 seconds for the medium case, and 0.01086 seconds for the large case. For tableau simplex, the corresponding runtimes were approximately 0.000254, 0.000386, and 0.000640 seconds. For revised simplex, they were approximately 0.000299, 0.000367, and 0.000604 seconds.

These measurements are small enough that all three methods run basically instantaneously. The timing comparison, therefore, verifies that the implementations run successfully on increasing model sizes.

Constraint activity. The small and medium cases are not capacity-constrained, so no shortage appears. The large case is capacity-constrained only in crewed mode. In the completed tableau and revised runs, the large-case crewed capacity slack is 0, while the dormant capacity slack is about 24.86 kW. This con-

firms that scarcity occurs in the active crewed regime rather than in the dormant keep-alive regime. The objective summary, runtime summary, runtime plot, and allocation plot are placed together at the end of the report in Tables 7–8 and Figures 1–2.

VIII. Discussion

The dataset progression shows where scarcity enters the model. Small and medium cases are fully supportable, so the shortage objective is zero. The large case adds ISRU demands large enough to bind the crewed source-capacity constraint. Once that happens, the shortage weights and delivery efficiencies determine which loads are protected first.

The large-case allocation comes from the weighted-service coefficients in Section V. Habitat operations have the largest coefficient and are served first. The rover loads follow, with the remote rover slightly discounted by its delivery efficiency. The science payload is then served. Ridge-side ISRU has a larger effective coefficient than crater-side ISRU, so the remaining crewed power goes to ridge-side ISRU before crater-side ISRU. This coefficient order explains why crewed crater-side ISRU is fully curtailed even though dormant-mode ISRU is fully served.

The agreement among PuLP/CBC, tableau simplex, and revised simplex supports the correctness of the implementation on the tested datasets. The strongest computational conclusion is qualitative and structural: the selected 50 kW crewed-mode scenario

supports habitat, mobility, science, and part of ridge-side ISRU, but it does not support full crewed ISRU operation. A larger source-side architecture, lower ISRU demand, different priority weights, or time-shifted ISRU operation would be required to remove the large-case crewed shortage.

IX. Limitations

The model does not include several realistic features in order to remain a continuous LP. No binary unit-commitment variables are included, so loads can be fractionally curtailed rather than switched on or off. Adding on/off commitment decisions would turn the application into a mixed-integer program. No battery state-of-charge or nonlinear storage dynamics are modeled, so the two operating modes are not connected by an intertemporal energy balance. No stochastic demand, generation uncertainty, outage process, or thermal-survival model is included. Finally, fixed delivery efficiencies replace a detailed electrical network with voltage, current, mass, fault-tolerance, and topology constraints. These simplifications make the model clear and solvable by simplex methods, but they limit the engineering interpretation of the numerical values.

X. Future work

Natural extensions would make the model more realistic but also more complex. A multi-period version could track storage state of charge across crewed and dormant intervals. A stochastic or robust version could represent solar availability, outages, and demand uncertainty. A mixed-integer version could require whole-load commitment for rovers, science payloads, and ISRU units. A network-flow or power-flow extension could replace fixed delivery efficiencies with explicit distribution constraints. Finally, sensitivity runs on penalty weights and source capacities would show how much additional power is needed before full crewed ISRU becomes feasible.

XI. Conclusion

This project formulates lunar base power allocation as a continuous linear program motivated by an Artemis Foundation Surface Habitat scenario. The LP separates hard habitat keep-alive demand from flexible habitat, mobility, science, and ISRU loads, then minimizes weighted kWh-shortage over crewed and dormant operating modes. The Part 1 simplex implementations were first checked on small general-form verification examples, then applied to three lunar datasets and compared with PuLP/CBC. The small and medium datasets are fully supportable. The large dataset is scarce only in crewed mode: habitat, mobility, science, and dormant ISRU are fully served, while crewed ISRU absorbs the shortage. The selected 50 kW crewed-mode source scenario is therefore adequate for the non-ISRU loads in the model, but not for full simultaneous crewed ISRU operation.

Acknowledgment of computational tools used

Python, NumPy, PuLP, CBC, and standard plotting tools were used for the standard-solver baseline. Python and NumPy were also used for the custom tableau simplex and revised simplex implementations. No standard solver was used inside the custom tableau simplex or revised simplex implementations.

Acknowledgment of Tools Used

Calculations, code structure, and report logic were checked using *Python*, *NumPy*, *PuLP/CBC*, and *ChatGPT (OpenAI)*. These tools were used to verify arithmetic, LP formulation consistency, runtime/result tables, figure values, and whether the custom tableau simplex and revised simplex outputs agreed with the standard solver baseline. These resources were employed as computational aids and to cross-check intermediate results for accuracy.

References

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End-of-report computational tables and figures

The following tables and figures collect the objective values, model-size comparison, runtime plot, and large-case allocation plot referenced in Section VII.

Table 7: Objective values, iteration counts, and interpretation.

Dataset	Solver objective	Tableau objective	Revised objective	Iters	Interpretation
Small	0.00000	0.00000	0.00000	6 / 6	All modeled loads fully served
Medium	0.00000	0.00000	0.00000	8 / 8	All modeled loads fully served
Large	239202.49536	239202.49263	239202.49263	11 / 11	Crewed-mode ISRU shortage appears

Table 8: Main computational results. Solver variables count model-level PuLP variables; simplex variables count the benchmark standard form.

Dataset	Solver vars	Simplex vars	Cons.	Tableau time	Revised time	Solver time
Small	18	20	14	0.0002543	0.0002989	0.0117556
Medium	24	26	18	0.0003860	0.0003672	0.0108813
Large	36	38	26	0.0006404	0.0006036	0.0108568

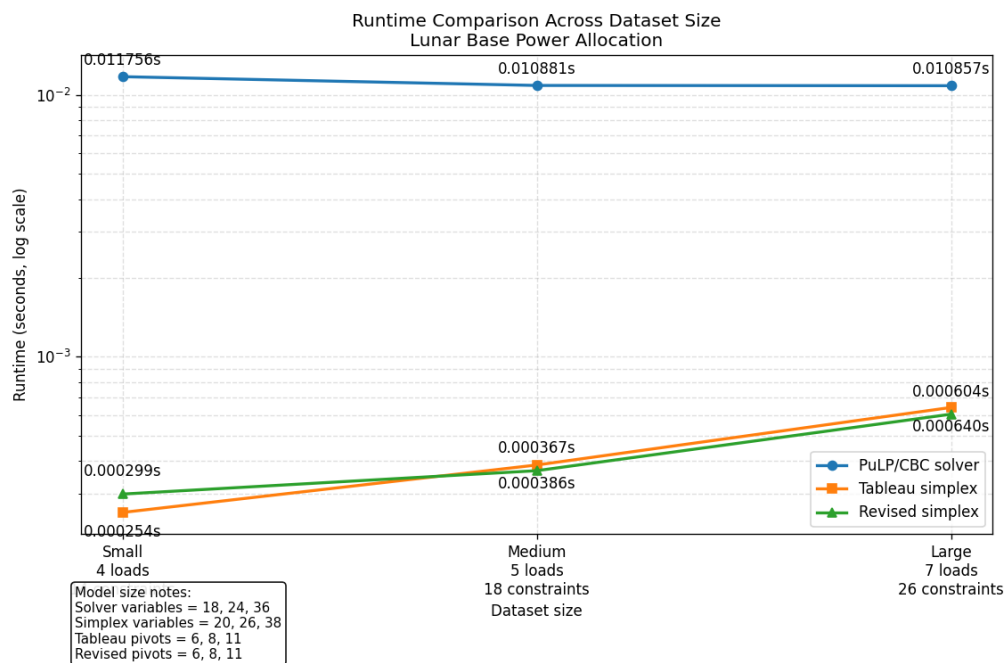


Figure 1: Runtime comparison across dataset size. Single-run wall-clock runtimes for tableau simplex, revised simplex, and PuLP/CBC across the small, medium, and large datasets. The Small dataset is cut off a little bit, it is 4 loads and 14 constraints. This can also be seen in Table 8 right above.

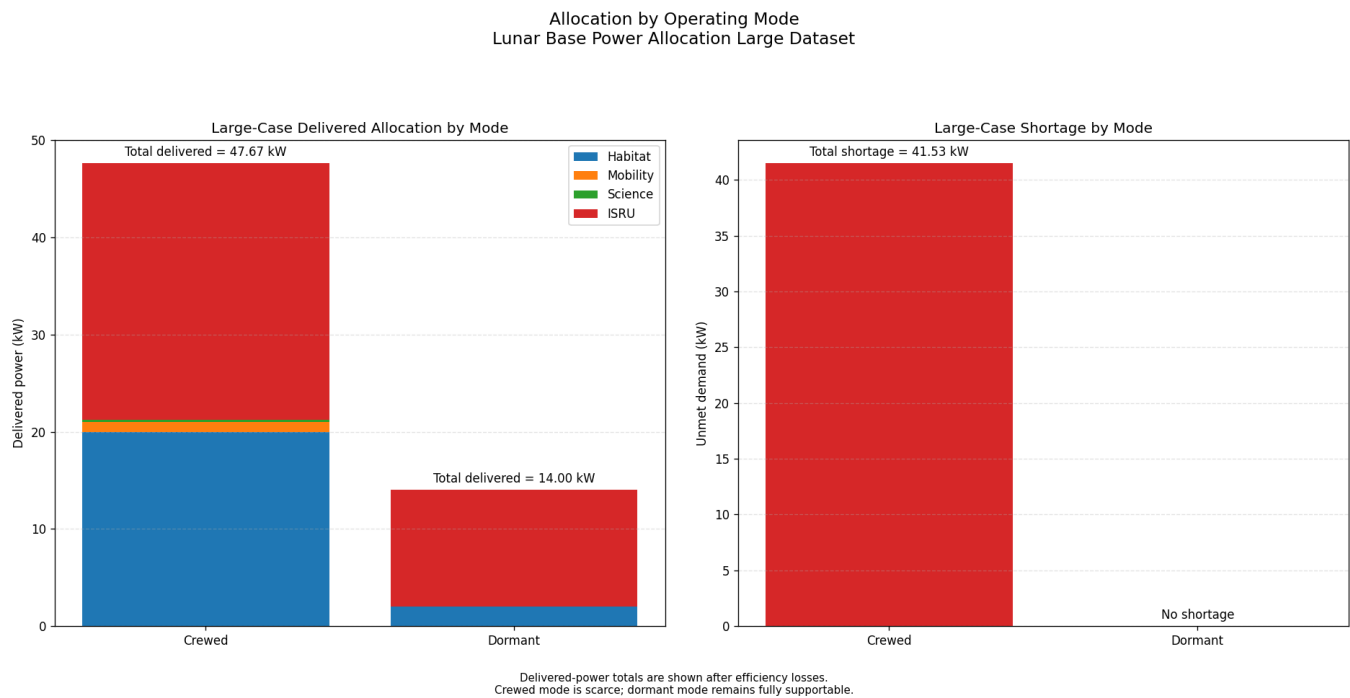


Figure 2: Delivered allocation and shortage by operating mode. Delivered power and unmet demand by operating mode for the large dataset. The crewed mode uses the fixed 2 kW habitat keep-alive load plus all 48 kW of flexible source-side capacity, while dormant mode fully serves the keep-alive ISRU loads and retains positive slack.

Appendix A. Part 1 verification output

The full console output for the Part 1 verification runs is included below. The output shows the exact method status, Phase I / Phase II pivot counts, standard-form sizes, and feasible solution values used to summarize Table 4. The appendix includes only the Part 1 verification output; the longer lunar run output is summarized in Tables 5–8 and Figures 1–2.

```
=====
SECTION 1: PART 1 GENERAL SIMPLEX IMPLEMENTATION CHECKS
=====
Feasible lower-bound LP
  Tableau | status = optimal | objective = 26.000000 | x = {'x1': 2.0, 'x2': 5.0}
          | std vars = 5 | std cons = 3 | phase1 = 3 | phase2 = 0 | runtime = 0.000333s
  Revised | status = optimal | objective = 26.000000 | x = {'x1': 2.0, 'x2': 5.0}
          | std vars = 5 | std cons = 3 | phase1 = 3 | phase2 = 0 | runtime = 0.000262s

Mixed-sense feasible LP
  Tableau | status = optimal | objective = 7.000000 | x = {'x1': 3.0, 'x2': 2.0}
          | std vars = 4 | std cons = 3 | phase1 = 2 | phase2 = 0 | runtime = 0.000128s
  Revised | status = optimal | objective = 7.000000 | x = {'x1': 3.0, 'x2': 2.0}
          | std vars = 4 | std cons = 3 | phase1 = 2 | phase2 = 0 | runtime = 0.000122s

Simple infeasible LP
  Tableau | status = infeasible | phase1 = 1 | phase2 = 0 | runtime = 0.000052s
  Revised | status = infeasible | phase1 = 1 | phase2 = 0 | runtime = 0.000057s
```